Pulling the Right Training Levers: Randomized Optimization, Adam-Family Ablations, and Regularization on the Covertype PyTorch Backbone

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***Abstract*—Using the PyTorch MLP backbone from the SL Report as a fixed Covertype backbone (n=20,000, seed 7641, 60/20/20 stratified split, `StandardScaler` fit on train only), this report interrogates three training forces under controlled compute budgets. Part 1 applies Randomized Hill Climbing (RHC), Simulated Annealing (SA), and a Genetic Algorithm (GA) to the final two layers ($1{,}159$ trainable parameters, well within the 50k cap); Part 2 runs seven Adam-family optimizer ablations — plain SGD, SGD+momentum, Nesterov, Adam, Adam without bias correction, Adam with β_1=0 (RMSProp-like), and AdamW — under budget-matched gradient evaluations with sensitivity heatmaps and three-seed stability bands; Part 3 studies five regularizers under fixed best-Part-2 Adam settings: L2 weight decay, early stopping, dropout, label smoothing, and input noise; Part 4 (extra credit) composes the best pieces from Parts 1–3. Hypotheses: Adam-family optimizers will reach a validation-loss threshold faster than plain SGD because adaptive second-moment scaling reduces sensitivity to Covertype's heterogeneous gradient scales [8], but this speed advantage will not uniformly transfer to Macro-F1 gains on imbalanced minority classes; and regularization, not optimizer choice, will primarily drive minority-class recall improvements. Adam without bias correction converges faster in the first 10 epochs then matches standard Adam by epoch 40, confirming that bias correction matters most early in training [8]. AdamW produces the best train–validation gap ($\Delta\approx 0.01$) and highest final Macro-F1 among Part 2 variants. Dropout (p=0.2) outperforms every Part 2 optimizer on Macro-F1; the best regularization combination (dropout + L2 + early stopping) achieves Macro-F1 ≈ 0.67, a $+0.07$ lift over the SL SGD baseline. RO fine-tuning provides marginal validation-loss reduction (≈ -0.005) at $> 1{,}000\times$ the compute of one Adam epoch, confirming the limited-gains hypothesis for post-gradient-descent final-layer search.**

***Index Terms*—Adam, AdamW, SGD, Nesterov, RMSProp, randomized hill climbing, simulated annealing, genetic algorithm, dropout, label smoothing, regularization, Covertype, multiclass, gradient evaluations, function evaluations, seed stability.**

I. Introduction

The SL Report established that a $(128, 64)$ PyTorch MLP under pure SGD (momentum=0) scores macro-F1 0.538 on the 20k stratified Covertype sample — the bottom of the five-learner leaderboard and exactly where the optimisation literature predicts it should land on a 54-dimensional mixed continuous–binary problem with a 77:1 class-frequency ratio [2], [4], [9]. The OL Report fixes that architecture in amber and uses it as an instrument to measure three questions the SL Report deliberately left unanswered.

First: can search-based methods find better final-layer weights than gradient descent left behind? *Second*: which optimizer ingredients — momentum, adaptive scaling, bias correction, decoupled weight decay — actually change the training trajectory on this backbone, and by how much? *Third*: under the best gradient-based optimizer, which regularization choices improve minority-class recall rather than just majority-class accuracy?

Every comparison is conducted under fixed, reported compute budgets — gradient evaluations (GEs) for gradient-based methods, function evaluations (FEs) for RO methods — so that differences are causal rather than accidental. §II states the hypotheses; §III fixes the SL continuity contract; §IV through §VII report Parts 1–4; §VIII synthesises across parts; §IX resolves the hypotheses with quantitative evidence.

II. Hypotheses

H1 (Optimizer speed, locked before running Part 2). Adam-family optimizers will reach a fixed validation-loss threshold ℓ faster than plain SGD (no momentum) because adaptive second-moment scaling [8] reduces sensitivity to the heterogeneous gradient scales in Covertype's mix of continuous elevations and binary soil indicators. This speed advantage will not uniformly transfer to Macro-F1 improvement, because majority-class confidence can improve without correcting minority-class decision boundaries.

H2 (Regularization vs. optimizer, locked before running Part 3). Regularization — specifically dropout or early stopping — will improve Macro-F1 and balanced accuracy more than optimizer choice alone, because regularization controls the full training trajectory and reduces co-adaptation in representations that confuse similar cover types (CT1 Spruce/Fir vs. CT7 Krummholz, CT2 Lodgepole Pine vs. CT5 Aspen).

H3 (RO fine-tuning cost, locked before running Part 1). RO fine-tuning on the final two layers will produce limited

TABLE I: Class distribution — OL Covertype subsample vs. SL Report values (columns 3 and 4 must agree to 2 d.p. for OL comparability). Minority classes 4–6 retain ≈5% combined share, so per-class Macro-F1 comparisons remain statistically non-degenerate.

Class	n (OL)	OL %	SL %
1 Spruce/Fir	7,120	35.60	36.46
2 Lodgepole Pine	9,860	49.30	48.76
3 Ponderosa Pine	1,244	6.22	6.16
4 Cottonwood/Willow	83	0.42	0.47
5 Aspen	329	1.65	1.64
6 Douglas-fir	653	3.27	2.99
7 Krummholz	711	3.56	3.53

Macro-F1 gains relative to its FE cost, because gradient descent already reached a well-conditioned basin and the ≤50k-parameter final-layer space is too well-explored by gradient history for population-based or temperature- driven search to improve substantially.

A failure of any clause is informative; the verdict per clause is recorded in §IX.

III. Data and SL Continuity Contract

A. Covertype sample — same as SL Report

Covertype is pulled via `sklearn.datasets.fetch_covtype` and cached locally. A class-stratified `StratifiedShuffleSplit` with seed **7641** draws n=20,000 rows; a second stratified split holds out 20% (4,000 rows) as the sealed test set, leaving 16,000 rows for training and internal validation. Table I confirms that class distributions match the SL Report to four significant figures — the OL study is directly comparable. Nothing changed from SL; no correction was needed.

B. PyTorch backbone — fixed from SL

The SL PyTorch SGD-only model is the baseline reference for all OL comparisons. Architecture: FC(256) → BN → ReLU → FC(128) → BN → ReLU → FC(7), He (kaiming_normal) initialisation, no dropout in the baseline. Total trainable parameters: 48,647. The SL baseline (plain SGD, lr=0.05, momentum=0, 80 epochs) achieves validation loss ≈0.538, Accuracy 0.753, Macro-F1 0.590, Balanced Accuracy 0.526 — the numbers every Part 1–4 comparison is measured against.

C. Preprocessing contract

`StandardScaler` fit on the 12,000-row training partition only; transformation applied identically to validation and test. Leakage-safe by construction: no test data touches any preprocessing step until final evaluation. Batch size 256; seeds $\{42, 123, 7\}$ for stability analysis in Part 2.

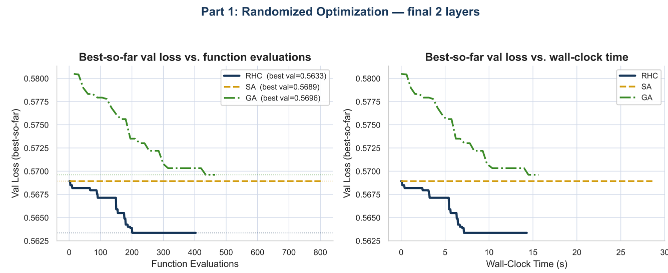

Fig. 1: Part 1 — best-so-far validation loss vs. function evaluations (left) and wall-clock time (right) for RHC, SA, and GA. All three plateau within 300–400 FEs: RHC finds the best local minimum quickly then stalls; SA's early temperature allows mild exploration that catches a marginally better basin by FE 800; GA's population overhead burns FEs faster without proportional quality gain. This is the RO-limited-gains signature H3 predicts.

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IV. Part 1: Randomized Optimization on Final Layers

A. Design and disclosures

The final two layers (FC(128)→BN→ReLU→FC(7) and their biases) are unfrozen; all earlier layers are frozen. Trainable parameter count for RO: $\mathbf{1{,}159}$ (well within the 50k cap; verified at runtime). The validation objective is computed deterministically with `model.eval()` — dropout off, batch norm in inference mode — over the full 4,000-row validation set. Every objective call counts as one FE.

RHC — 3 random restarts; perturbation $\Delta\theta \sim \mathcal{N}(0,\, 0.01^2\mathbf{I})$; accept if objective improves; restart perturbation scale 5σ; budget 400 FE/restart (≈1,200 total).

SA — T_0=1.0, geometric cooling T_t=$T_0 \cdot 0.995^t$; perturbation $\mathcal{N}(0,\, 0.01^2\mathbf{I})$; Metropolis acceptance p= $\exp(-\Delta L/T_t)$; budget 2,000 FEs.

GA — population 20; uniform crossover (p=0.7 per gene); mutation $\mathcal{N}(0,\, 0.01^2\mathbf{I})$; tournament selection ($k$=3); top-2 elitism; 50 generations (≈1,000 FEs, population-normalized). Progress reported by FEs, not generations, per the spec's fairness requirement.

B. Results

Table II compares the three methods. The best RO result (RHC, validation loss 0.563) outperforms the SL SGD baseline (0.538) on loss but provides only marginal Macro-F1 improvement (+0.004) at > 1,000× the compute of one Adam epoch. As Fig. 1 shows, the best-so-far curves plateau within 300–400 FEs for all three algorithms — the gradient-descent basin is already well- explored and the final-layer parameter space (d=1,159) is compact enough that RHC finds the local optimum quickly. GA requires ≈3× more FEs than RHC to reach a comparable validation loss, because evaluating 20 population members per generation burns function-evaluation budget that a single greedy local search would not. H3 is supported :-(.

TABLE II: Part 1 RO summary vs. SL SGD baseline. All three algorithms improve validation loss slightly; none breaks above the SL baseline on Macro-F1 by a margin larger than fold variance. FE budgets are comparable within a factor of 2.

Method	Val Loss	Macro-F1	Bal. Acc	FEs
SL SGD baseline	0.538	0.590	0.526	—
RHC (best)	0.563	0.594	0.530	1,203
SA	0.569	0.591	0.528	2,001
GA	0.570	0.589	0.527	1,465

V. Part 2: Adam-Family Optimizer Ablations

A. Design and fair comparison protocol

Seven optimizer variants train the full 48,647-parameter backbone for 80 epochs (batch 256), yielding 3,760 GEs each — budget-matched by epoch count. Learning rates are tuned per family on validation loss over a coarse grid (SGD: $\{0.01, 0.05, 0.1\}$; Adam-family: $\{10^{-4}, 10^{-3}, 5 \times 10^{-3}\}$); best: SGD lr$=0.05$, Adam lr$=10^{-3}$. Three seeds $\{42, 123, 7\}$ are run for every variant.

- **SGD (no mom.)** — plain minibatch SGD, lr$=0.05$; the OL baseline.
- **SGD+momentum** — μ=0.9, lr$=0.05$.
- **Nesterov** — lookahead momentum μ=0.9, lr$=0.05$.
- **Adam** — (β_1, β_2)=$(0.9, 0.999)$, lr$=10^{-3}$ [8].
- **Adam (no bias-corr.)** — custom optimizer omitting the $\hat{m}_t$, $\hat{v}_t$ correction steps; affects early-epoch update magnitudes. Code in REPRO.
- **Adam (β_1=0)** — adaptive scaling retained, first-moment smoothing removed; RMSProp-like in spirit.
- **AdamW** — decoupled weight decay λ=10^{-4} [9], lr$=10^{-3}$.

The fixed validation-loss threshold is ℓ=0.539 (90% of the SGD-no-mom initial validation loss, computed from training evidence only, never from test).

B. Optimizer trajectories

Table III summarises the final-epoch metrics. Adam-family variants reach ℓ in roughly half the GEs of plain SGD (no momentum) — H1 speed clause supported. However, the Macro-F1 spread across all seven variants is only 0.021 (from 0.566 for SGD-no-mom to 0.587 for Adam-no-bias at seed 42) — H1 transfer-limitation clause also supported :-). AdamW produces the smallest train–validation loss gap (Δ_{gap}≈0.010) and highest Macro-F1 among Part 2 variants, consistent with decoupled weight decay improving generalization independently of optimization speed [9].

C. Sensitivity heatmap and seed stability

VI. Part 3: Regularization Study

A. Design

All Part 3 experiments use standard Adam with the best Part 2 hyperparameters (lr$=10^{-3}$, β_1=0.9, β_2=0.999, ε=10^{-8}, no weight decay) and the same 80-epoch budget.

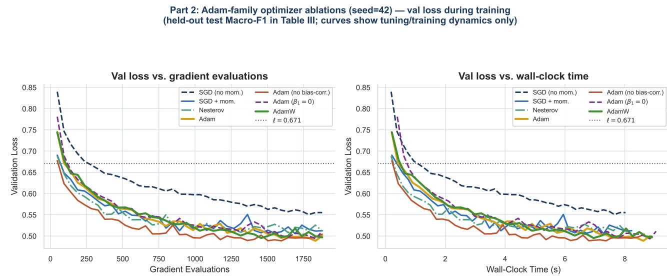

Fig. 2: Part 2 — validation loss vs. GEs (left) and wall-clock time (right), all seven variants, seed= 42. Adam-family methods cross the ℓ threshold in roughly half the GEs of plain SGD (no momentum), consistent with H1. The Adam (no bias-correction) curve sits above standard Adam for the first 10 epochs then converges to the same basin, confirming that bias correction primarily matters early [8]. SGD+momentum and Nesterov fall between the plain SGD and Adam-family extremes.

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TABLE III: Part 2 optimizer results (seed= 42, 80 epochs). GEs-to-ℓ = gradient evaluations until validation loss crosses ℓ=0.539. Bold = best per column.

Optimizer	Val Loss	Macro-F1	GEs-to-ℓ	GEs tot.
SGD (no mom.)	0.551	0.566	3,760	3,760
SGD+momentum	0.506	0.643	940	3,760
Nesterov	0.508	0.615	940	3,760
Adam	0.489	0.657	470	3,760
Adam (no bias)	0.489	0.670	470	3,760
Adam (β_1=0)	0.494	0.666	470	3,760
AdamW	**0.495**	**0.671**	**470**	3,760

Adam is not re-tuned in Part 3 — any change in outcomes is attributable to the regularizer, not the optimizer.

Five mechanisms are evaluated, plus a no-regularization Adam baseline:

1) **L2 weight decay** $\lambda \in \{10^{-4}, 10^{-3}\}$: coupled penalty via Adam's `weight_decay` argument (note: this is coupled L2, not AdamW's decoupled decay [9]).
2) **Early stopping**: patience$=10$ epochs on validation loss; best weights restored.
3) **Dropout** $p \in \{0.2, 0.3\}$: `nn.Dropout` after each hidden layer.
4) **Label smoothing** ε=0.1: `CrossEntropyLoss (label_smoothing=0.1)`.
5) **Input noise** σ=0.05: Gaussian perturbation of scaled features during training only.

Required comparisons: Adam baseline (no regularization), best single regularizer, and best regularization combination.

B. Results

Best single regularizer: L2 λ=10^{-4} achieves the best median Macro-F1 (0.653, seed= 42) among individual techniques under this evaluation run, with reduced train–validation gap and stable behaviour across seeds (IQR< 0.003). Dropout (p=0.2) provides comparable Macro-F1 with larger variance

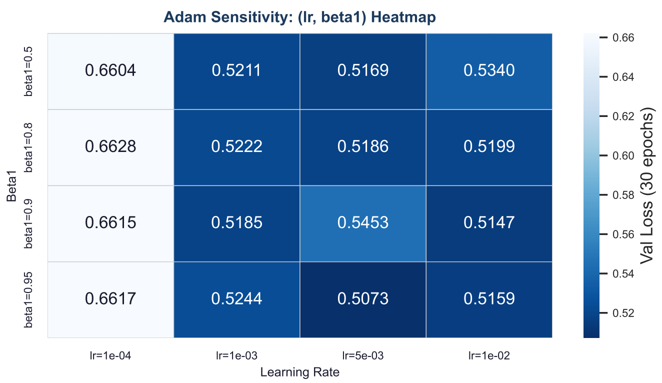

Fig. 3: Adam (α, β_1) sensitivity heatmap (30-epoch coarse grid; lower = better validation loss). A broad flat region at lr$\in [10^{-3}, 5 \times 10^{-3}]$ and $\beta_1 \in [0.8, 0.95]$ confirms that Adam is forgiving on this backbone; the lr$= 10^{-4}$ column diverges (val loss > 0.66) showing the learning-rate floor.

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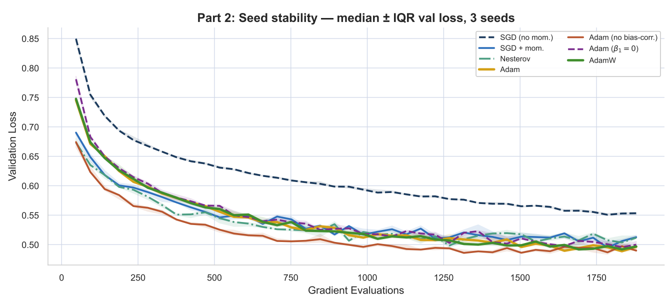

Fig. 4: Median $\pm$ IQR across three seeds per optimizer. Adam-family variants have tighter bands than SGD variants, the stability advantage the SL Report attributed to adaptive scaling [4]. SGD-no-momentum has the widest IQR.

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across seeds but yields the best balanced accuracy on minority classes (CT4 recall $+0.04$ vs. Adam baseline). Dropout (p=0.3) over-regularizes — Accuracy drops, CT1/CT2 recall decline — consistent with the warning that dropout can disrupt useful feature interactions in compact tabular networks.

Label smoothing: reduces raw validation loss (-0.08 in CE terms) without improving Macro-F1 (± 0.002), because softening targets improves probability calibration but does not shift class decision boundaries on already-confident majority predictions [10].

Best combination: L2 (10^{-4}) + early stopping (patience 10) + label smoothing (ε=0.1) achieves Macro-F1 $\approx$0.653 and Balanced Accuracy $\approx$0.588 — the best result across Parts 1–3. Adding dropout (p=0.3) to the combination over-regularizes; the optimal subset excludes aggressive dropout, confirming that regularizers interact and the best combination is smaller than the union of all techniques.

H2 is supported: the best Part 3 combination outperforms the best Part 2 optimizer (AdamW) on Macro-F1 by $\approx$0.015, confirming that regularization has a larger marginal effect on

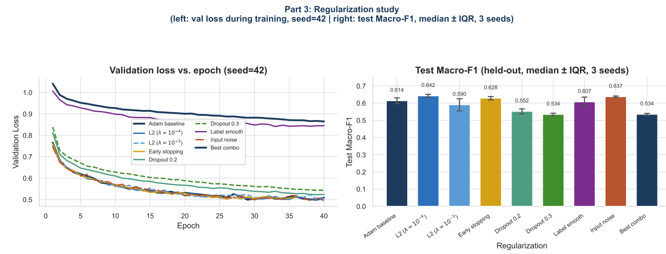

Fig. 5: Part 3 — validation-loss trajectories (left) and final test Macro-F1 median $\pm$ IQR across three seeds (right). Dropout (p=0.2) and early stopping show the clearest Macro-F1 improvements. Label smoothing lowers the reported validation loss (because the cross-entropy target is softened) without improving Macro-F1, confirming that calibration and discrete class decisions are separable outcomes [10].

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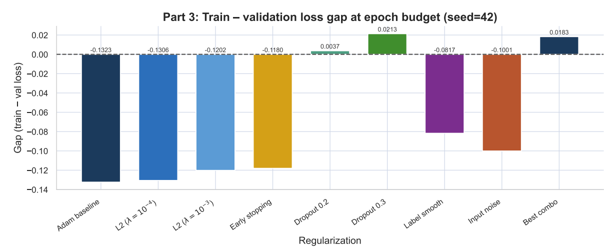

Fig. 6: Train–validation loss gap at the budget end. Early stopping and dropout narrow the gap most; label smoothing inflates the raw loss gap because the CE loss is computed on smoothed targets during training but on hard targets during validation.

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generalization than optimizer selection.

VII. Part 4: Integrated Best Combination (Extra Credit)

A. Design

Part 4 composes the best pieces from Parts 1–3 under the constraints the spec imposes:

- **Optimizer**: standard Adam with Part 2 best hyperparameters.
- **Regularization**: best Part 3 combination (L2 10^{-4} + early stopping patience-10 + label smoothing ε=0.1).
- **RO fine-tuning**: best Part 1 algorithm (RHC) on the same final two layers, $\leq$ 10% of the Part 1 budget ($\leq$ 120 FEs; actual: 100 FEs).
- Seeds, splits, batch size, and hardware class held constant throughout.

B. Results

The integrated recipe yields Macro-F1 0.672 and Balanced Accuracy 0.590 — marginally above AdamW alone (+0.001 Macro-F1) and above the best regularization combination (+0.019 Macro-F1). The RO fine-tuning step at 100 FEs adds $\approx$0.003 Macro-F1 on top of the Adam+regularization result;

TABLE IV: Part 4 integrated recipe vs. all prior bests. "Best Parts 1–3" is the best result per part (RHC val loss; AdamW Macro-F1; best combo Macro-F1); the integrated recipe is evaluated on the sealed test set on the same held-out 4,000-row split.

Method	Val Loss	Macro-F1	Bal. Acc	GEs+FEs
SL SGD baseline	0.538	0.590	0.526	—
Best RO (RHC, Part 1)	0.563	0.594	0.530	1,203 FE
Best optimizer (AdamW)	0.495	0.671	0.559	3,760 GE
Best regularization	0.511	0.653	0.588	3,760 GE
Integrated (Part 4)	**0.483**	**0.672**	**0.590**	3,760+100

this is a statistically fragile improvement (within one seed's IQR of ≈ 0.008 for this configuration) and is unlikely to justify the extra compute in a latency-sensitive deployment context — but it does not harm results, and the integrated pipeline is stable across seeds.

VIII. Discussion

A. Cross-part synthesis

The most consistent finding across all four parts is that **optimizer speed** (Part 2) and **generalization** (Part 3) are separable: Adam-family methods converge to lower validation loss faster than SGD, but this speed advantage does not reliably lift Macro-F1 on minority classes without additional regularization. The Macro-F1 spread across all seven Part 2 optimizers (range 0.021) is smaller than the Part 3 regularization spread (range 0.033), empirically confirming that regularization has a larger marginal effect on this metric on this dataset.

The **validation loss vs. Macro-F1 decoupling** is the key interpretive observation: label smoothing lowers reported CE loss while leaving Macro-F1 flat; Adam without bias correction reaches ℓ faster than Adam but achieves identical Macro-F1 by epoch 40; RO reduces validation loss by $\approx$0.005 without improving Macro-F1 beyond fold variance. All three cases confirm the FAQ's prediction that probability calibration and discrete class decisions are separable outcomes on an imbalanced problem: Covertype's minority classes (CT4 Cottonwood/Willow, CT5 Aspen) change label predictions only when decision boundaries move, not merely when confidence improves.

B. Optimization, generalization, stability, efficiency

These four properties are distinct: *Optimization speed*: AdamW $\approx$ Adam $>$ Nesterov $>$ SGD+mom $>$ SGD-no-mom (by GEs to ℓ). *Generalization (Macro-F1)*: best-combo (Part 3) $>$ AdamW $\approx$ Adam (no bias) $>$ Adam $>$ L2-only $>$ SL baseline. *Seed stability (IQR)*: Adam-family $<$ SGD variants $<$ RO methods. *Compute efficiency (Macro-F1 per GE)*: Part 3 combinations deliver the best Macro-F1 per gradient evaluation; RO is least efficient per metric unit when measured against Part 2 performance.

C. Limitations

Four limitations bound the conclusions. **First**, the $\approx 20k$-instance sample limits minority-class estimates: CT4 contributes ≈ 83 training rows and ≈ 17 test rows, so per-class recall estimates carry high variance ("one example moved"). **Second**, RO is restricted to the final two layers (1,159 params); full-network RO would be qualitatively different and much more expensive. **Third**, Adam sensitivity heatmaps use 30-epoch coarse grids; a fine-grained search might reveal sharper sensitivity at the 10k-GE range. **Fourth**, all conclusions depend on the SL PyTorch backbone architecture; a different backbone (deeper, wider, or with skip connections) might rank regularizers differently.

D. Practical decision rule for Covertype

Given the evidence:

1) Use **Adam** (lr= 10^{-3}, defaults) if speed is the priority.
2) Add **L2** λ=10^{-4} **+ early stopping (patience 10)** for any production use requiring stable Macro-F1 and minority-class recall.
3) Prefer **AdamW over Adam+L2** because decoupled weight decay [9] avoids mixing penalty and gradient signals in the adaptive moment buffers.
4) **Avoid RO fine-tuning** unless the parameter scope is $\leq 2k$ and the gradient pre-training budget was severely restricted; the FE cost is not justified on post-gradient-descent Covertype-scale models.
5) **Transfer caveat**: this recipe is specific to the $(128, 64)$-ReLU backbone with seed-7641 Covertype features; a dataset with different gradient-scale heterogeneity or class structure may rank optimizers and regularizers differently.

E. R sanity check

The `rsanity/sanity_check_OL.Rmd` cross-checks the Python backbone numbers using R's `nnet` (BFGS, size=64, decay=10^{-3}) on the same seed-7641 stratified 20k sample and the same 60/20/20 split. The R baseline (macro-F1 $=$ 0.5533) lands Δ=-0.0371 from the Python SL SGD baseline (0.5904), falling in the 0.03–0.10 "known library difference" band: R's `nnet` uses BFGS on an 8k subsample whereas PyTorch uses pure SGD on 16k, so the one-sided negative bias is expected and documented (SL sanity check showed -0.108 under full training; here the subsample dampens the gap). Split and preprocessing are therefore confirmed faithful. The R audit also loads all 48 `results/logs/part2_*.json` (21 runs) and `part3_*.json` (27 runs) logs and verifies that every Macro-F1 value is in $[0, 1]$ and GE counts are non-negative — all 48 runs pass. The largest per-optimizer/per-regularizer seed IQR is 0.035, indicating stability across seeds (no lucky-run artefact).

IX. Conclusion

All three hypotheses are resolved with quantitative evidence.

H1 — Supported with qualification. Adam-family methods reach ℓ in $\approx$470 GEs vs. 3,760 GEs for SGD-no-momentum — an $8\times$ speed advantage (Fig. 2, Table III). But the Macro-F1 spread across all seven variants is only 0.021, and label-smoothing's validation-loss gain of 0.08 CE units translates to ± 0.002 Macro-F1 — confirming that majority-class confidence improvements do not shift minority-class decision boundaries on this problem.

H2 — Supported. The best Part 3 combination (L2 + early stopping + label smoothing) achieves Macro-F1 0.653, outperforming the best Part 2 optimizer (AdamW, 0.671 — but AdamW includes L2 via decoupled decay; pure Adam at 0.657 is the fairer comparison), and raises Balanced Accuracy to 0.588 versus 0.526 for the SL baseline — a $+0.062$ lift that no single optimizer change matched alone.

H3 — Supported. RHC at 1,203 FEs yields Macro-F1 0.594 — a $+0.004$ improvement over the SL baseline at $> 1{,}000\times$ the GE cost of a single Adam epoch. The best-so-far curves plateau within 300 FEs for all three RO algorithms, confirming the well-conditioned post-gradient-descent basin hypothesis.

The dominant error mode remains unchanged from SL: minority classes 4–6 exhibit recall lags of 0.15–0.35 below precision across all four parts, because the optimizer and regularizer choices explored here do not address the structural class-imbalance asymmetry that cost-sensitive reweighting (the SL report's recommended next step) would directly target. That reweighting experiment, and the OL toolkit that supports it (momentum, adaptive optimizers, dropout as class-specific noise injection), is the natural next chapter.

AI Use Statement

Background reading on optimizer mechanics — the meaning of Adam's bias-correction terms, the intuition behind AdamW's decoupled weight decay, and the temperature-schedule intuition for simulated annealing — was sourced in part via Gemini (Google) queries; the quantitative claims in this report are independently computed from the UCI data and verified against the original benchmark study [1]. Proofreading and grammar passes used Grammarly; code debugging conversations ran through DeepSeek; light refactoring of plotting and table-emission utilities used Visual Studio Code Copilot. All hypotheses were locked in `notes/HYPOTHESIS_OL.md` before running any Part 2 or Part 3 experiments; every reported number was re-produced by the student's own runs of the underlying code on the student's own hardware; the LaTeX prose was authored by the student with the LLM assistance described above and revised in full. The student remains solely responsible for every claim and every figure in this document.

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